How to Practice

On the homework you actually do, this class averages **94%**. On the quizzes covering the same material, **70%**. That gap is not about effort, and it is not about who is "a math person." It is about *conditions*. You are practicing under conditions that never repeat on a quiz.

What the homework does for you that a quiz will not

- **It names the method.** Every set covers one section, so the heading has already chosen between the Product Rule and the Chain Rule before you read the problem.
- **It lets you be wrong for free.** Unlimited attempts mean a wrong answer costs nothing and you still finish at 100%.
- **It leaves everything open** — your notes, the book, a worked example two problems up.

A quiz does none of the three. So practice by taking them away.

Four habits

1. **Start on a blank page.** Close the notes, close the book, close the worked example. Write the problem at the top and go. Check only once you are done or genuinely stuck. If you read an example first, what you practiced was *copying*.
2. **Practice choosing, not just doing.** Collect ten problems from different sections, cover the headings, and write only the *first step* of each. You are not solving — you are deciding. The ungraded **Mixed Practice — which technique?** set in Canvas does this for you, unlimited, and tells you why after every answer.
3. **Know where each rule comes from.** Quoting $\frac{d}{dx}\arcsin x$ is not the same as deriving it from $\sin y = x$. Quiz 6 asked for that derivation and many of us had only memorized the result. Once a week, derive from scratch: tan, cot, sec, csc from sin and cos; arcsin and arctan from their triangles; a^x by rewriting it as $e^{x \ln a}$.
4. **Spread it out, and start cold.** Twenty minutes on four days beats eighty minutes the night before. Open each session by re-deriving yesterday's rules on blank paper, *before* you look at anything.

The four days before an exam

1. Go down the **"What You Should Be Able to Do"** list that closes every section. Read one line, close the book, and do a problem for it.
2. Take the **practice exam** closed-book, timed, in one sitting. Do not look at the solutions.
3. **Grade it yourself.** Beside every error write *why*: wrong method, right method with an algebra slip, or did not know the rule. Those three need three different fixes, and only you can tell them apart.
4. **Redo only what you got wrong** — from a blank page, not by rereading your corrections.

Feels like studying. Is not.
Rereading the notes · watching a worked example and following along · redoing problems you already got right · opening the solution "just to get started" · **watching an AI work it** · highlighting.
Each one feels smooth, and smooth is the problem: the work is being done by the page in front of you instead of by you. A quiz is the first time you find out.

A note on AI, because it is the fastest way into that trap. Asked for a solution it gives you a flawless one, and every step looks obvious — because someone else chose it. Choosing the steps was the assignment. Use it the way this course asks: *after* you have an answer, to check it, never before, to get one. The test is the same one on the *Using AI in This Course* page — if you could not reproduce the work on paper, alone, in fifteen minutes, it is not yours yet.